Angles & Triangles

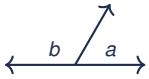
NC.8.G.5 • Angle relationships, parallel lines, triangles, similarity

Unit 4 • Front

1 Angle Pairs



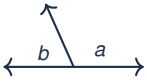
Vertical: $a = b$



Linear pair: $a + b = 180°$

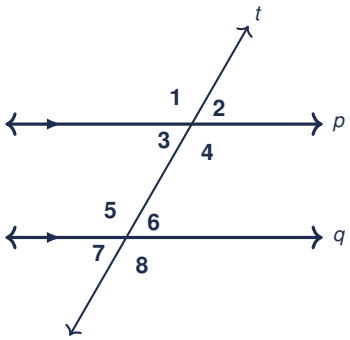


Complementary: $a + b = 90°$



Supplementary: $a + b = 180°$

2 Parallel Lines Cut by a Transversal



$p \parallel q$ (arrows show parallel lines)
t = transversal

Only 2 sizes! If $\angle 1 = 110°$:
$\angle 1 = \angle 4 = \angle 5 = \angle 8 = 110°$
$\angle 2 = \angle 3 = \angle 6 = \angle 7 = 70°$

Pair	Where	Example	Rule
Corresponding	same corner at each line	$\angle 1$ & $\angle 5$	**equal**
Alternate interior	inside, opposite sides of t	$\angle 3$ & $\angle 6$	**equal**
Alternate exterior	outside, opposite sides of t	$\angle 1$ & $\angle 8$	**equal**
Same-side interior	inside, same side of t	$\angle 3$ & $\angle 5$	**add to** 180°
Same-side exterior	outside, same side of t	$\angle 1$ & $\angle 7$	**add to** 180°

Equal pair: alt. interior
$\angle 3 = 3x + 10$, $\angle 6 = 4x - 15$

$$\begin{aligned} 3x + 10 &= 4x - 15 \\ \underline{-3x} & \quad \underline{-3x} \\ 10 &= x - 15 \\ \underline{+15} & \quad \underline{+15} \\ 25 &= x \end{aligned}$$

Each angle: $3(25) + 10 = 85°$

Add-to-180 pair: same-side int.
$\angle 3 = 2x + 20$, $\angle 5 = 3x + 10$
$(2x + 20) + (3x + 10) = 180$

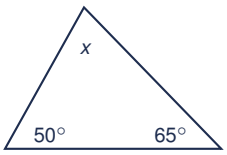
$$\begin{aligned} 5x + 30 &= 180 \\ \underline{-30} & \quad \underline{-30} \\ 5x &= 150 \\ \frac{5x}{5} &= \frac{150}{5} \\ x &= 30 \end{aligned}$$

$$\begin{array}{r} 30 \\ 5\overline{)150} \\ -15 \\ \hline 0 \\ -0 \\ \hline 0 \end{array}$$

Angles: 80° and 100°. $80 + 100 = 180$ ✓

3 Triangle Angles

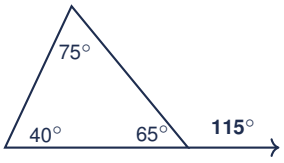
Interior angles add to 180°**.**



$$\begin{aligned} 50 + 65 + x &= 180 \\ 115 + x &= 180 \\ \underline{-115} & \quad \underline{-115} \\ x &= 65° \end{aligned}$$

Why? Tear off the 3 corners. They line up to make a straight line (180°).

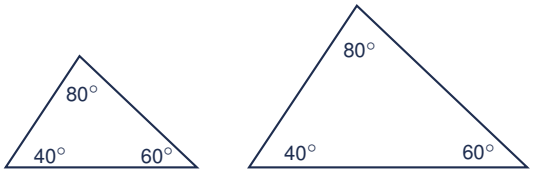
Exterior angle = sum of the 2 far (remote) interior angles.



$40 + 75 = \mathbf{115}°$

Exterior + its neighbor = linear pair:
$115 + 65 = 180$ ✓

4 AA Similarity: Two Matching Angles



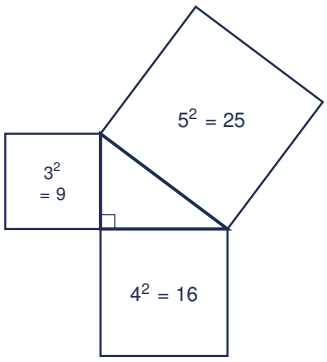
1. Two angles match (40° and 60°).
2. The third must match too: $180 - 40 - 60 = 80°$.

So the triangles are SIMILAR: same shape, can be a different size.

Pythagorean Theorem

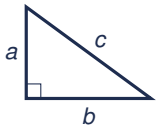
NC.8.G.6 • NC.8.G.7 • NC.8.G.8

5 The Theorem: Only for RIGHT Triangles



$9 + 16 = 25$ The two small squares fill the big one.

$$a^2 + b^2 = c^2$$



a, b = **legs** (make the right angle)

c = **hypotenuse**: across from the right angle. Always the **longest** side.

Common triples: 3, 4, 5 6, 8, 10 5, 12, 13 8, 15, 17

6 Find the Missing Side

Missing hypotenuse: ADD

Legs 6 and 8

$6^2 + 8^2 = c^2$
$36 + 64 = c^2$
$100 = c^2$
$\sqrt{100} = c$
10 $= c$

Missing leg: SUBTRACT

Leg 5, hypotenuse 13

$5^2 + b^2 = 13^2$
$25 + b^2 = 169$
$\underline{-25} \quad \underline{-25}$
$b^2 = 144$
$\sqrt{b^2} = \sqrt{144}$
$b =$ **12**

Not a perfect square? Estimate.

Legs 2 and 5: $4 + 25 = c^2$, $c = \sqrt{29}$

1. $25 < 29 < 36$: roots 5 and 6
2. $\frac{29 - 25}{36 - 25} = \frac{4}{11}$
3. $4 \div 11 \approx 0.36$
4. $5 + 0.36$

$\boxed{c \approx 5.4}$

$$\begin{array}{r} 0.3\,6 \\ 11\overline{)\,4.0\,0} \\ -\,3\,3 \\ \hline 7\,0 \\ -\,6\,6 \\ \hline 4 \end{array}$$

7 The Converse: Is It a Right Triangle?

1. Longest side is c.
2. Plug into $a^2 + b^2 = c^2$.
3. Equal? **YES**, right triangle.
   Not equal? **NO**.

Sides 9, 12, 15

$9^2 + 12^2 \stackrel{?}{=} 15^2$
$81 + 144 \stackrel{?}{=} 225$
$225 = 225$ ✓
YES, right.

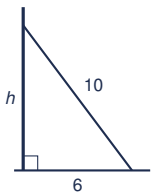
$$\begin{array}{r} 1\,5 \\ \times\ 1\,5 \\ \hline 7\,5 \\ +\ 1\,5\,0 \\ \hline 2\,2\,5 \end{array}$$

Sides 5, 7, 9

$5^2 + 7^2 \stackrel{?}{=} 9^2$
$25 + 49 \stackrel{?}{=} 81$
$74 \neq 81$
NO, not right.

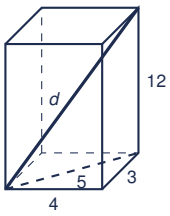
8 Real World, 3D, and the Coordinate Plane

Ladder: A 10 ft ladder. Base is 6 ft from the wall. How high does it reach?



$6^2 + h^2 = 10^2$
$36 + h^2 = 100$
$h^2 = 64$
$h =$ **8** ft

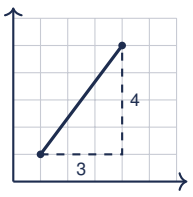
3D box: 3 by 4 by 12. Find the diagonal d.



Floor:
$3^2 + 4^2 = 25$
so floor diagonal = 5

Box:
$5^2 + 12^2 = 169$
$d = \sqrt{169} =$ **13**

Distance: (1, 1) to (4, 5)



No distance formula needed!

1. Draw the right triangle.
2. Count legs: 3, 4
3. $9 + 16 = 25$
   $d = \sqrt{25} =$ **5**